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# ON THE COLLECTION OF EDUCATION STATISTICS IN DEVELOPING COUNTRIES: PURPOSE, PRINCIPLES, PROCEDURES AND PROBLEMS

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## 1. INTRODUCTION

As a consequence of a vigorous policy in favour of education, most developing countries could, at the threshold of the 1980s, look back on two decades characterized by a growth in pupil numbers unparalleled in history. From being a comparatively small sector, concerning chiefly the lives of a relatively small urban middle and upper class, education has in the span of one generation developed into a large-scale operation, affecting the conditions of life of most sections of the population. This unprecedented expansion of educational systems has placed very heavy demands on the administrators and planners of the education sector. Despite efforts to cope with these demands, the development of national administrative, planning and research capacities often lags behind the growth in size and complexity of the education enterprise.

A prerequisite for improving developing countries' administrative and planning capacities in the field of education is often to improve their capacities to *collect* and *analyze* statistical data related to the activities of this sector. The lack of information on key aspects of educational development and the often poor quality of available information have become growing concerns of both producers and users of education statistics.

There are many reasons for the present increased realization and appreciation of the importance of accurate, detailed and timely education statistics. The sheer size and complexity of the education sector is one. Another factor of growing importance is the fact that the education sector since the mid-1970s has found itself in a tightening financial squeeze between rigid budget ceilings and rapidly rising costs. On the *budget* side, there is reason to believe that the days when an increase in the share of national income devoted to the education sector could be obtained quite easily may be but a memory. Although there is no obvious ceiling for the proportion of public budgets a government can spend on education, in practice each nation has thresholds beyond which the burden of educational expenditures may seriously cripple the supply of other public services. The end to the preferential treatment enjoyed by education during the past two decades is also related to the present waning confidence in education as the prime mover of development and the growing levels of 'educated' unemployment in many countries.

As regards the *cost* side, education remains an extremely labour-intensive sector. As the large number of young teachers recruited in the developing countries during the last two decades stay on in the system, emolument will increase since teachers' pay scales normally include automatic increases based on length of service and provide for grade increases of teachers improving their qualifications. As the squeeze between budget ceilings and costs grows tighter, many countries will also be faced with difficult choices as regards the priority to be given to expansion of enrolment at different levels of education since the unit cost generally increases steeply by increasing levels of education.

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In short, the need for efficient utilization of available resources is becoming more important than ever. Apart from the above-mentioned tightening financial squeeze and the many unsolved education problems in developing nations, wise spending of public resources is a duty in its own right. And a *necessary* condition for achieving improvements in this area is, for a good number of developing nations, an improved data base. This is not to say that this is a *sufficient* condition; statistics alone cannot solve any problem. However, properly collected and analyzed statistics can help administrators and planners to use better existing educational resources both with respect to efficiency (i.e. doing things more rightly) and effectivity (i.e. doing the right things). The major role of statistics is precisely to assist decision-makers by providing the background information needed for informed decisions.

The capacity of developing countries to collect and analyze education statistics varies considerably. While some countries have well-developed statistical systems, others have only recently started to collect education statistics in a systematic manner. Most countries fall between these two extremes, i.e. they collect education statistics from schools on a routine basis but nevertheless experience serious problems, either as regards the quality, detail and timeliness of the data collected or as regards their capacity to analyze these data.

The purpose of the present paper is to discuss some general aspects of data gathering in the field of education. The paper starts by highlighting the main purposes of education statistics. Section 3 discusses some basic principles that education statistics should fulfil. Section 4 reviews six approaches used to a varying degree for the collection of statistics in the field of education while Section 5 treats some special problems experienced in the collection of education statistics in many developing countries.

The paper is meant as a general introduction to the purposes, principles, procedures and problems of collecting education statistics in developing countries. The questions of what type of statistics to collect and how to go about the collection and analysis of these statistics are not dealt with here.

## 2. PURPOSE OF EDUCATION STATISTICS

Education statistics are no end in themselves. They are tools which help us to monitor current developments within the education sector as well as to understand how this complex sector works in order to be able to plan for the future. We may, admittedly somewhat arbitrarily, distinquish between the following six interrelated and overlapping functions that education statistics shall serve:

(i) *The descriptive function*

The statistics shall describe the current education situation in terms of purpose, pupils, teachers, administrative and other personnel, physical accommodation, material and equipment available as well as the end product in terms of school leavers, graduates and research.

(ii) *The comparative function*

The statistics shall permit comparisons of educational provision and performance between different institutions, regions, population groups, etc. within a country as well as with educational systems in other countries. Both the value and the many pitfalls of such comparisons should be stressed.

(iii) *The monitoring and accounting function*

The education sector is a large-scale operation that may account for as much as 20–30% of the total public budget. It produces the nation's qualified manpower and recruits in turn a large part of this manpower as students, teachers, administrators, etc. Education is furthermore a highly decentralized activity with a large local and voluntary element. The need for day-to-day administration, monitoring and control of such a complex sector is obvious. To perform these functions well requires a substantial amount of detailed statistics.

(iv) *The planning function*

Educational planning provides a bridge between the analysis of the current situation and the formulation of policies and programmes for the future. If educational planning is to serve policy-making processes effectively by providing information on likely results of alternative choices, planners must have at their disposal a comprehensive set of reliable and timely statistics.

(v) *The evaluating function*

A variety of statistics is required for assessing the performance of the educational system against stated objectives. Fundamentally, there are two types of questions that one is concerned with in making evaluations. The first is whether or not a particular educational programme or action was implemented according to its stated guidelines ('process evaluation'), the second is the extent to which the programme or action made a change in the desired direction ('impact evaluation'). Both types of evaluation, carried out independently or together, are essential. Unfortunately, very little evaluation of any type is carried out in the education sector due, partly, to lack of statistical data.

(vi) *Educational research or the theory-generating function*

This function is naturally closely related to the planning and evaluating functions. Based on accurate and adequately defined and collected education statistics, causal relationships between factors inside as well as outside the educational system may be tested. There is a very urgent need for such research in developing countries, both to study the relationship between educational inputs and outputs (e.g. the effect of teachers, material and equipment on pupils' performance, attitudes and aspirations) and to study the relationship between education and the society at large (e.g. the effect of the methods of financing on the distribution of educational benefits, the contribution of education to changes in people's attitude to development, effects of education on the income distribution).

Traditionally, the stress in data gathering has been on the descriptive, comparative and monitoring functions rather than on the planning, evaluating and research functions. This is a direct consequence of the fact that statistical systems in developing countries have evolved largely as by-products of administration, often derived from systems set up by former colonial powers. The importance of educational planning, evaluation and research is however well recognized in developing countries and the urgency of establishing a data base of sufficient detail, accuracy and timeliness to serve these purposes is becoming increasingly clear. However, to establish such a data base requires the creation of a well-organized data-gathering machinery. This takes time and there is still a long way to go in most developing nations before this will be achieved.

The present situation as regards education statistics in developing nations may be illustrated with a quotation from a survey on educational planning in Asia conducted in 1975 by the UNESCO Regional Office for Education in Asia and Oceania. In commenting on the status of 'research, data and information' the survey states that:

> Despite two decades of educational planning, the educational planners have still to contend with inadequacies in data gathering, storing and retrieving mechanisms. The adequacy, accuracy and quality of data are generally referred to [by countries in the Asian region] as major constraints. The data requirements for planning are now of a different order because of the increasing sophistication of planning techniques, but there has been no corresponding improvement in the data collection machinery. Thus the data collected prove, more often than not, inadequate and even irrelevant.
>
> (UNESCO, 1975, p. vi)

The paucity of basic data is considered a serious obstacle to a planned development of education by national planners. For example, in the case of Bangladesh, Qadir and Ahmed note that:

> Lack of accurate information on various aspects of primary education was considered since long a major hindrance to planning education to meet the country's need. In the absent of sufficient research and educational planning, together with lack of reliable and adequate data, the policies and programmes in the past had to be based on preconceived notions which had little or no relevance to the social and economic realities of the country.
>
> (Quadir and Ahmed, 1980, p. 1)

Similar observations are made by a senior educational planner in Thailand commenting upon problems of data availability in his country:

> Despite the fact that Thailand devotes more attention to educational statistics than many developing countries, the planners — both at the central level and in the Provinces and Regions — are still confronted with substantial statistical problems.
>
> (Tajaroensuk, 1975, p. 135)

Similar observations are made for African countries in the Terminal Report prepared by a former regional adviser on educational statistics in Africa:

> . . . The shortage of adequate and reliable data is in many developing countries so drastic that work on educational planning and investment projects is seriously hampered.
>
> (UNESCO, 1977, p. 2)

Further:

> As a general judgement about the work situation in education statistics in most of the countries, it can be said that, although some laudable efforts have been made here and there, the data collection is very fragmentary, and the analysis and publication of the data leave much to be desired.
>
> (UNESCO, 1977, p. 9)

Finally, we quote a report on a seminar on data collection in education in the Caribbean, organized by the Caribbean Community Secretariat. In its discussion of 'problems or concerns demanding consideration' the report says that:

> (a) data-collecting has been narrowly conceived or neglected altogether;
> (b) useful data are not always available and available data is not always useful. . .
>
> (Caribbean Community Secretariat, 1978, p. 26)

The report goes on to comment on the lack of dissemination of existing statistics and the lack of co-ordination at various levels in developing data collecting systems. We shall return to these aspects in Section 5.

The first step in improving the data base is to clarify the purposes for which the data gathered will be used. Like most other types of statistical information, education statistics are expensive to collect. It is therefore uneconomical to collect statistics without a reasonably clear idea of the purpose for which they are to be used. This point is of crucial importance. As explained in the following section, one of the main criteria for data collection is that the information shall be 'user-related'. This can only be achieved through a close co-operation between the education statisticians, planners, administrators and other user groups.

## 3. PRINCIPLES FOR COLLECTION OF EDUCATION STATISTICS

The practical problems encountered when employing the education statistics available in developing countries for the purpose of planning and administration are many, and they vary between countries, types of statistic and uses to which the statistics are put. Notwithstanding such differences, some problems are common to most countries, types of statistic and uses. For example, data are often incomplete, inaccurate or both. They are furthermore often out of date and too aggregated. Some types of statistics essential for planning purposes (e.g. data on pupil and teacher flows, cost data) are often missing, and so on.

We shall list below some general principles that should guide the decision-maker as regards which data to collect and how to collect them. It will be clear from studying this list that the principles included are not mutually exclusive. Furthermore, they compete with each other. No set of data can be perfect, one has always to weigh improvements in one dimension against sacrifices in another. For example, to increase the number of elements on which data are collected may, for the same level of resources, reduce the accuracy of the data collected and/or increase the time-lag between collection and publication. Gains in accuracy may imply increases in costs as would most other improvements required in the data presently available in developing countries. The trade-offs between the various principles listed must therefore be decided for each country taking due account of national conditions and constraints. (The order in which the items are included in the following list appear does not indicate order of importance.)

(i) *Data should be based on unambiguous definitions*

Although this might at first sight appear to be a requirement too obvious to deserve inclusion in this list, it is unfortunately true that this basic principle of data gathering often is not sufficiently observed. It is therefore necessary to stress at the outset the overriding importance of developing clear and unambiguous definitions for all elements on which information is to be collected.

The pitfalls are many in this area. Terms which at first appear to be quite clearly defined may on reflection be rather vague. Suffice it to say here that there sometimes is doubt about the meaning of even basic terms such as enrolment, repeaters, promotees and wastage. It is useful in this connection to distinguish between two aspects, i.e. cases where the definition of a

given term varies between countries and cases where different suppliers and/or users of data within a given country interpret national definitions differently.

As regards the first aspect, we should like to stress the importance of a standardized statistical terminology for the purpose of international comparisons in the field of education. Although the planning and administration of education are very largely purely national concerns, the problems encountered in this field are often similar in nature in many countries, and much can therefore be learned from studying other countries' experiences and politics. However for such international comparisons of educational developments and policies to be worthwhile they must be based on common terms of reference. An essential aspect of such common terms of reference is that any statistics that are used must have, as nearly as possible, the same meaning in the different countries involved.

Even when data are used for purely national purposes, to employ the terms proposed in international recommendations concerning educational statistics will in most cases present definite advantages. For example, many of the methods of educational planning most commonly employed are based on the use of educational statistics collected according to these definitions. This is for instance the case for the simple pupil flow models used for reconstructing cohort flows and for projecting school enrolment. These models imply a particular way of defining new entrants, repeaters, promotees and dropouts. In a case where these flows are defined otherwise, this may imply that the results drawn from such models cannot be interpreted in the usual way. As an example, we mention that some countries define 'repeaters' as 'pupils who fail the examination at the end of a given school year' (i.e. 'the failures') instead of as 'pupils who at the beginning of two consecutive school-years are enrolled in the same grade'. If this is the case, the meaning of the results calculated for cohort flow indicators of the type normally derived for pupil flow models (e.g. survival by grade, graduation and wastage) will not be as described in textbooks presenting such models.

With respect to the second aspect mentioned above, i.e. different suppliers and/or users of data within a given country interpreting national definitions differently, this happens quite frequently. In particular, those who are the primary suppliers of education statistics, i.e. educational institutions, may not always have the time, resources and trained personnel required to give the completion of the annual questionnaires the necessary attention. The questionnaire itself may moreover not be sufficiently clear, the definitions of terms may not be precise enough, and so on.

(ii) *Data should be complete*

There are at least two ways in which the educational statistics available in developing countries is often incomplete. Firstly, the *coverage* may be incomplete in the sense that not all pupils, teachers, etc. are included. One common cause of incomplete data is that some schools are not included on the list of schools to which questionnaires are sent. Incomplete school-lists is a problem common to many developing countries. There are many reasons for schools not being included on the school lists. Public education is often provided by many ministries as well as by a wide array of government agencies, and all of these activities may not be known to those responsible for data collection. Furthermore, privately managed education plays a significant role in most developing nations. There is also the problem of unrecognized schools, i.e. institutions which, for various reasons, are not recognized by the Ministry of Education[1].

Another serious concern in many developing countries is the fact that some schools, although included on the school list, do not complete the questionnaires.

The second way in which data are often incomplete in developing nations is the *lack of certain key data elements* absolutely essential to carrying out even the most rudimentary planning of educational activities. As an example of such data we mention statistics on pupil flows in countries where the level of repetition and/or dropout is significant, and the very common paucity of data on education costs. We shall return to this problem under point (vi) below.

Lack of key data and incomplete coverage of existing data are serious obstacles to efficient planning and monitoring of educational systems in developing nations. For example,

how can a country's progress towards targets such as universal primary education be assessed if, due to incomplete coverage of data, we do not know how many children are attending school? How can future school enrolment, teacher requirements and costs be projected if one does not know how many of the children of school-going age have already entered school?

As an example of the problem lack of key data poses for educational planners, we quote the following statement made by a national planner concerning the situation in Thailand in the mid-seventies. In commenting on 'unavailability of important data' the report states:

> The major statistical problem faced by planners during the preparation of the Third Five-Year Plan (covering the period 1971-1976) was the unavailability of large amounts of essential data directly connected with educational systems as well as other data needed for planning. As far as data directly connected with education are concerned there was almost a complete lack of any data in the following fields: (1) institutional unit costs; (2) sources of finance; (3) availability of physical facilities; (4) utilization of physical facilities; (5) age and standard of existing facilities; (6) wastage through student drop-out and repetition; (7) field of specialization of teachers; (8) proportion of newly qualified teacher trainees becoming teachers.
>
> (Tajaroensuk, 1975, p. 136)

The report goes on to discuss how the lack of these key data items complicates the planning process. The data items included in this quotation cover the aspects on which statistics are most generally lacking in developing countries.

Finally, we note that the problem of incomplete data is further complicated when there are yearly variations in the coverage of the data which are available due, for example to fluctuations in the response rate. In the case of data on enrolment, this may cause considerable variations in the flow rates.

(iii) *Data should be accurate*

Although this is a rather obvious basic requirement, the issue of accuracy is not as clearcut as it might seem. Few, if any, sets of education statistics need to be one hundred percent accurate; indeed, it would in most cases be an unwise use of scarce resources to aim at such a high level of precision. The level of accuracy needed for planning purposes moreover differs between different types of information. For example, while it would be essential to achieve a high level of accuracy for the data collected on the *number* of pupils and teachers, a lower level of precision might be acceptable for statistics pertaining to certain pupil and teacher characteristics (e.g. age, socioeconomic background, causes of dropout). Every effort should be made to achieve a high level of accuracy for the minimum set of data collected through the annual school census. For other types of data needed on an *ad hoc* basis to study specific educational issues, a lower level of precision might sometimes be acceptable. For the sake of greater speed and economy, it is often possible in such cases to employ estimates based on secondary sources rather than on direct measurement.

After having established an adequate machinery for conducting the annual school censuses, the first step towards ensuring accuracy is to identify and guard against possible biases in reporting. For example, if grants or teacher assignments are based upon enrolment, this may be an incentive for schools to inflate their enrolment figures. Attendance may likewise be exaggerated because low attendance rates may reflect badly upon teachers and administrators. Apart from more or less deliberate misreporting of this type, a large number of sources of errors exist. The problem of ambiguous definitions of terms has already been mentioned above. Other types of error result from inaccuracy in recording of the information, from adding up the questionnaires as well as from copying the statistics from one form to another. The importance of scrutiny and correction of errors at all stages in the collection and compilation of statistics cannot be overemphasized. All too often the usefulness of data which are collected at great cost is severely impaired by carelessness especially in copying and checking.

(iv) *Data should be sufficiently disaggregated*

At the outset data are collected at the school level and are later on aggregated. Aggregation means loss of information. Frequently, the only data readily accessible to planners, research and policy-makers refer to the nation as a whole or, at best, to geographical or administrative regions. For many types of studies such data are too aggregated, e.g. studies of the impact of school input variables (i.e. teachers, instruc-

tional material, class size, etc.) on the pupils' performance.

Ideally it should be possible to relate one variable to another at least at the school level. This enhances greatly the utility of the statistics. Such flexibility is however not obtained free of cost, and it requires that data be gathered and compiled in such a way that they can be used for a number of different purposes.

(v) *Data should be available on a timely basis*

A problem common to many developing countries is a quite long time period between the date at which education statistics are collected and the date at which they are made available to the users. Illustrations of this rather sad state of affairs can be found in education development plans of many countries where, in order to have an enrolment figure for the year the plan was established, one has often had to project two or three years ahead of the latest year for which data were available. This means that in order to prepare an enrolment projection to be included in, say, a five-year plan one actually needs to project six or seven years ahead of the latest observed figure. For example, in the case of Thailand, Tajaroensuk states that:

> At the time when the detailed projections for the Third Five-Year Plan were made (during 1970), the latest complete school census available was that for 1967. Thus the projection for the period 1971-1976 sometimes had to be based on data no more recent than 1967 . . .
>
> (Tajaroensuk, 1975, p. 136)

It is obvious that this increases considerably (and unnecessarily) the uncertainty of the projection.

Lack of up-to-date statistics is a serious concern to planners and policy-makers. How can one monitor current education trends if it takes years to know what these trends are? In general, a long time period between collection and publication greatly reduces the utility of the data for administrative and planning purposes. In the worst case data made available long after the period for which they are gathered may become obsolete before they ever reach the planner or decision-maker. This would mean that the considerable amount of scarce human and financial resources spent on collecting, compiling and publishing the data are to a large extent wasted.

To speed up the process will in general mean actions at the school level to encourage quicker response to the questionnaires, more trained clerical staff at the regional and central level to verify, rectify, compile and consolidate the data and more equipment in terms of calculators, duplicators, etc. The point here is that this would generally be a quite small investment compared to the resources already used directly and indirectly in collecting the data.

One remedy often proposed in order to reduce the time-lag between collection and publication is to use computers for the processing of the data once they reach the education statistics unit. We shall discuss this point briefly in Section 5. Suffice it to say here that, although the use of computers presents several advantages once the data are ready for processing, most of the delay in making education statistics available to the users is not due to long processing time, but to the time it takes to collect, verify and rectify the data.

Timeliness is clearly in competition with some of the criteria already mentioned above, e.g. completeness, degree of detail and, to some extent, accuracy. It is often preferable for the planners and policy-makers to have data which are somewhat imperfect in terms of completeness, detail and accuracy when they need them rather than better data when it is too late.

(vi) *Data should be user-related*

A quite common experience among users of statistical information is that data tend to be compiled in a way that prevents meaningful analyses of issues which require the combination of several sets of data. An important example is budget and expenditure data which are usually categorized by item (e.g. teacher salaries, textbooks, chalk) rather than by function (teaching, administration, research) or by type and level of education. This makes it very difficult, if not impossible, to relate these statistics to other sets of data to study aspects such as various types of unit cost (e.g. teaching cost per pupil, administration cost per pupil or school, average pupil cost at different levels of education). It is even difficult to calculate basic measures such as pupil-teacher ratios by level of education in countries where teachers are engaged in teaching at, e.g. both first and second level education.

The above problem may be illustrated by a quotation from an Indian study. In a section commenting on the difficulty of linking expenditure data to other types of data the report states:

> . . . It is also difficult to relate expenditure to enrolments. While enrolments are given by stages and schools, expenditure is reported by institutions. An institution is designated by the highest class it teaches; many institutions have more than one stage. A high school may have primary, middle and high school classes; it is, therefore, difficult to determine precisely the per student cost of education at various levels of education.
>
> (Saraf *et al.*, 1980, p. 180)

In order to organize the data in a form that maximizes their potential utility for the users, it is necessary for those responsible for gathering the statistics to have a fairly clear idea of what uses these statistics might be put to. As explained in Section 2 time and effort spent on clarifying these questions at an early state in the data-gathering process make it possible to avoid collecting (at great cost) information which later on proves to be ill-suited to analytical purposes. To ensure that the data collected are user-related is a two-way process in which administrators as well as planners and statisticians need to cooperate closely.

(vii) *Data should be accessible*

All too often data collected at great cost are practically inaccessible to many user groups. In the worst case (and such cases exist) the information on pupils, teachers, etc., is not even extracted from the questionnaire returns received from the schools. A more common situation is that the information is compiled and summary tabulations are prepared and used for planning purposes within the Ministry of Education. However, these data are not necessarily published in a form that makes them available to a wide audience. Even when a publication is prepared, it sometimes takes a very long time to appear and is issued in a rather limited number of copies. Lack of resources and low priority for education statistics publications within the government printshop are two common reasons for such delays. For example, in the case of India, Saraf *et al.* say that:

> The data having been processed, is prepared in manuscript form for printing and is sent to the Government of India Press. The press usually takes more than a year to supply printed copies of the publication as it has to cater to the needs of all the Central Ministries, Parliament and other Government organizations.
>
> (Saraf *et al.*, 1980, p. 177)

This completes our brief review of general principles for collection of education statistics. In the following section we shall mention some principles for data-gathering procedures.

## 4. PROCEDURES FOR COLLECTING EDUCATION STATISTICS

*Ways of obtaining education statistics*

There are essentially six ways in which educational statistics can be obtained. They are not mutually exclusive and a well-organized data collection programme will utilise all six. They are:

(i) Regular censuses of all educational institutions;
(ii) Regular sample surveys covering selected data items;
(iii) *Ad hoc* sample surveys to provide information on topics of special interest;
(iv) Population censuses;
(v) Sample surveys drawn from the total population, usually in conjunction with population censuses;
(vi) Routine reporting of data obtained as a by-product of educational administration.

We shall below comment on each of these six ways of obtaining education statistics. Two points should, however, be stressed at the outset. Firstly, not all of the above six ways of data collection are equally relevant for all types of education statistics. For example, while *ad hoc* sample surveys are very useful for collecting data needed for special studies (e.g. on factors such as causes of drop-out and repetition, reasons for pupils not attending school regularly, socioeconomic background of pupils, work aspiration of school leavers), data on other items (e.g. on teachers, pupils, finance) need to be collected on a regular basis through censuses covering all institutions.

Secondly, whichever of the six methods is used in obtaining the data, there are two basic approaches to the collection and processing of the statistics. The first is based on *aggregated* statistics on data items such as pupils, teachers, various types of educational facilities, finance,

etc., supplied by the basic reporting unit, usually the educational institution. The statistics are aggregated in the sense that they cover the *total* number of pupils enrolled in each grade, the *total* number of teachers in the school, etc.

The second approach is *individual* reporting where institutions, instead of submitting aggregated data for the entire institution, submit information on each component of the aggregate. This implies that each individual covered is given a unique code number on which all information referring to this individual is recorded. The reporting can be the result of answers to questionnaires filled out by the individual himself or by the responsible authority (e.g. the school). This form of data collection is generally referred to as an individualized data system (ID system). The main distinction between such a system and a system which is not based on individualized data is that in the case of an ID system the permanent individual characteristics of pupils and teachers (e.g. sex, year of birth, background data) would need to be recorded only once (i.e. when the pupil or teacher first entered the institution) and all matching of current educational activity with previous experience and with individual characteristics can be done based on this recording. In the case of a system not based on individual data all the data required on individual characteristics of pupils and teachers must be especially recorded each time it is required.

Among the main advantages of data collected through an ID system as compared to aggregate data is the elimination of double counting of teachers and pupils. The system is particularly well suited to the collection of flow statistics on these two main data elements. Accurate and detailed flow statistics can be derived easily, and the storage of the statistics is flexible and permits quick production of cross-tabulations. It is, however, clear that to derive maximum advantage from such a system requires the use of computers.

Notwithstanding the above advantages, only a very limited number of industrialized countries have so far introduced an ID system for education statistics. No developing country has installed such a system. There are many reasons for choosing to rely on the more traditional system of aggregated statistics. In addition to the very high initial costs of establishing an ID system, developing countries often lack the equipment and trained manpower essential for operating such a system. Furthermore, to be certain that the investment is worthwhile, a data collection procedure must be established which is certain to produce reasonably reliable base data. No useful purpose is served by installing a costly system for storing unreliable data. There is also in many countries considerable resistance to the idea of a centralized individual register on pupils. Moreover, decentralized statistics are required in a decentralized educational system and special care would have to be taken to ensure an efficient and rapid feedback of information from the central register to the various levels of local administration as well as to the individual institutions.

Thus, the use of an ID system for collection of education statistics in developing countries is very much a question for the future. This aspect will, therefore, not be further discussed in this paper.

(i) *Regular censuses of educational institutions.* Regular censuses of educational institutions form the backbone of the education data collection system of practically every country. To carry out such a census normally implies that the person responsible for administering each institution (e.g. the headmaster or headteacher of each school) answers a number of questions concerning his institution. For the most part these questions would deal with the pupils and teachers in the institution, but they can also concern equipment, finance, school buildings and other facilities. Usually the information is collected through questionnaires. The carrying out of questionnaire surveys for the purpose of collecting education statistics gives rise to many questions which cannot be discussed here.

Unlike population censuses which normally are carried out once every tenth year and which cover the whole population enumerated at one particular date, censuses of educational institutions are normally carried out annually, their coverage is limited to the educational system and different data elements may be enumerated at different points in time (e.g. pupils at the beginning of the year and the expenditures at the end). The date at which the census is held is

an important aspect since the number of pupils and teachers normally changes during the school year. It is generally recommended that the data on pupils and teachers should refer to the beginning of the school year. By 'beginning' is normally meant a month or so after the first school day of the school year. This is particularly important for the enumeration of pupils in countries where school attendance is not strictly compulsory and where some of the children who turn up the first days of the school year may, for various reasons, quit school after a short period. If the count refers to a date after the school year has started, these early dropouts may be omitted from the register. Inclusion in school registers of pupils who do not attend school at all (apart from perhaps a few days) occurs in many developing countries. Such errors are significant in many countries. They lead to overestimation of enrolment and may cause considerable distortion in the estimate of the promotion, repetition and dropout rates.

A related phenomenon common to many developing countries is high dropout during the school year, i.e. pupils who do attend several months before leaving. Such intrayear dropout is particularly found in the first grade of primary education. In cases where intrayear dropout does take place, it implies that the enrolment figures reported by the school are affected by the date chosen for the count as well as by the procedure followed for exluding from the register pupils who do not attend school regularly. In order to cope with this problem many countries (particularly in Latin America) count their pupils more than once during the school year.

As already mentioned, the regular census of educational institutions forms the backbone of the education data collection system in all countries. Although the other five ways of collecting data mentioned above may provide useful, and in some cases, indispensable supplementary information, these other methods can in no way replace the annual school census.

(ii) and (iii) *Regular and* ad hoc *sample surveys.* Data collected through sample surveys are subject to three main types of errors:

(a) *Sampling error:* However efficient the selection of the sample is, it is unlikely to replicate exactly the characteristics of the total population. For a random sample it is theoretically possible, by making certain assumptions, to calculate the size of the sampling error associated with a given sample size as well as the probability of the occurence of the error. This means that the estimate of the sampling error can be used to determine the size of the sample required to keep this error within the limits found acceptable to those performing the investigation.

(b) *Survey error:* This refers to errors such as incorrect counting by the investigator, incorrect responses by the persons interviewed, errors in recording of the responses, errors in the transcription, computation and compilation of the data, etc. This is often an important source of error, particularly in developing nations.

(c) *Time-sampling error:* This type of error arises from the uneven occurrence of events over time. For example, in the field of education, the results derived from a sample investigation to determine enrolment will, due to intrayear dropout and irregular attendance, depend on the time during the school year at which the investigation is carried out.

Sampling techniques are quite extensively used in developing countries for collecting statistics in fields such as demography, employment, agriculture and health. So far such techniques have however not been much used in the field of education. This is unfortunate since many types of education statistics could be fully or partly collected through sample investigations. For example, commenting upon the cost-effectiveness of various data-gathering techniques, McMeekin says that:

> . . . One means of reducing data-gathering costs is through sample studies.
>
> The reliability of sample studies (given certain conditions) is not generally understood by non statisticians. Few administrators of developing educational systems recognize that a well-designed sample survey can produce policy information that is virtually as accurate as a one-hundred percent census; and that sampling can improve the manageability of data by avoiding data glut, reducing costs, and increasing speed of analysis. The net result is that it is often possible to obtain useful policy information more effectively through sampling than through full surveys . . .
>
> (McMeekin, 1975, p. 64)

For example, sample surveys could be used to:

(a) check the accuracy of data collected through the annual school census;
(b) collect statistics on elements on which information is needed for *ad hoc* studies and for formulating policy on particular topics;
(c) collect statistics on items on which data are not needed annually and which, consequently, should not form part of the annual school questionnaire;
(d) collect detailed background information of a type that can best be obtained by the use of trained investigators rather than through mail questionnaires.

The use of sampling techniques in the field of education should be regarded as a way of supplementing the information collected through the annual school census, which will remain the backbone of the education data-gathering system. However, to benefit from this approach developing countries need to train statisticians and investigators in sample survey techniques and procedures. The UNESCO Office of Statistics is concurrently promoting studies in this area.

(iv) and (v) *Population census and sample surveys drawn from the total population.* While information on the current situation in the school system is collected mainly from educational institutions, information on the educational characteristics of the population at large is obtained from general population censuses and surveys associated with such censuses. The population census questions related to education generally cover literacy, educational attainment and school attendance. As we in this report are mainly concerned with current educational statistics, collected directly from institutions through annual school census, we shall not discuss further problems related to the collection of data on cducational attainment and literacy.

(vi) *Data that emerge as a by-product from administrative needs.* As stated previously, statistical systems in developing countries have evolved largely as a product of educational administration. Administrative records constitute the main source of information on important aspects such as educational finance and expenditures, teacher salaries and qualifications, scholarships, examination results, etc.

The need for administrative purposes of the above types of statistics is obvious. Education is a large-scale operation accounting for as much as 20–30% of public budgets in developing countries. Detailed financial statistics are therefore required for budgeting and accounting purposes. Moreover, to monitor educational standards as well as the output of the educational system in terms of school-leavers and graduates requires data on examination results. The Ministry of Education is moreover one of the largest employers in any country and quite detailed personnel records must necessarily be kept for administrative purposes such as control of the payment, of enrolments, for fixing salary scales, for promotion, appointment policies, etc.

Faced with this situation whereby a large part of the statistics needed for planning, evaluation and research purposes are contained in files and registers designed solely for administrative purposes, the task of the educational statisticians and planners is *not* to attempt to control these statistics, but to ensure they are designed and stored in such a way that they can also serve non-administrative purposes.

In closing this section we would like to stress that in choosing between the various possible procedures of data gathering discussed above, the question of costs is crucial. Just as cost-effectiveness techniques should be applied to education programmes, they should similarly guide the selection between alternative ways of collecting a given set of statistics. If proper cost-benefit studies were performed one would probably find that a significant part of the data presently collected through the school censuses could have been collected both more accurately and at lower costs through sample surveys.

## 5. PROBLEMS FACED BY DEVELOPING COUNTRIES IN IMPROVING THEIR DATA BASE

Section 3 discussed some general principles that education statistics should meet. As pointed out in that section, many developng countries are not always able to adhere strictly to these principles and their data base consequently suffers from a number of shortcom-

ings. We shall in the present section highlight some special problems faced by those in charge of education statistics in developing countries.

(i) *Lack of trained staff at all levels*

The reason most commonly given to explain shortcomings in the statistics available on education in developing countries is lack of trained staff. There is a severe shortage of qualified statisticians working in this field. Equally important is the shortage of assistants and clerks who have undergone at least a brief training programme in basic statistical notions and techniques. For example, summarizing his experience from visiting nine African countries the erstwhile Regional Adviser on Education Statistics in Africa states that:

> In most of the countries visited the unit responsible for educational statistics has a very limited and insufficiently trained staff. In the extreme case, only one official is working in the whole area of data collection, compilation, analysis and publication of school statistics.
>
> (UNESCO, 1977, p. 8)

The lack of trained staff is felt at all levels of data gathering. It is in fact important to keep in mind that, with the exception of countries with comparatively small populations, the questionnaires submitted by schools pass through one or more local/provincial/regional/state administration(s) before reaching the unit responsible for education statistics at the national level[2]. The work of these local administrations is often decisive in determining the quality in terms of accuracy, completeness and timeliness of the education statistics collected. It is in many cases no exaggeration to say that while the extent to which the data are actually used for planning purposes largely depends on the competence of the staff at the central level, the quality of the data largely depends on the availability of trained inspectors, planners and administrators at the local level. This is so because many of the errors commonly found in education statistics are committed at the source (e.g. reporting errors made by the schools) or during the compilation and consolidation of the questionnaires at the local and provincial levels. These kinds of mistake are difficult both to trace and correct at the central level. Improved staffing at the lower levels of the data collection hierarchy, combined with training of those completing the questionnaires at the school level are, therefore, two necessary conditions for improving the quality of education statistics in developing countries.

As an illustration of the lack of trained staff at the local level in some countries we quote the following statement from a survey of the problems of collecting education statistics carried out by the Pakistani authorities. Reporting on the situation in Punjab it is said that:

> At the district level, there is no such person who is well trained for the purpose of collection of educational statistics. To talk of a trained statistical hand at district level is an imaginary proposition, at present in all the district of Punjab there is not even a single person who is exclusively meant for the job of collection of true educational statistics . . .
>
> (Bureau of Educational Planning and Management, 1976, p. 1)

With respect to Baluchistan the report states that:

> No routine [of collecting education statistics] has as yet been established, because of inadequacy of staff.
>
> (Bureau of Educational Management, 1976, p. 7)

Among the main reasons for this lack of officers trained in the collection and analysis of education statistics, we mention the general shortage of trained statisticians in many developing nations, the low priority given to statistics in general and to education statistics in particular, and the tendency of trained education statisticians to transfer to more prestigious fields of statistics such as economic statistics. We shall return to these aspects below.

(ii) *Low priority and prestige of education statistics*

Traditionally education statistics, as well as statistics on other social concerns, have been given a much lower priority in developing countries than have various types of economic statistics. This stress on economic statistics is no doubt partly a consequence of the prevailing development strategy of the post-war era. This strategy has been centered around economic growth through industrialization and it has not always paid sufficient attention to various social and distributional aspects of the growth process. This being said, the strategy did give a high priority to the education sector as the main provider of qualified manpower. In view of this fact and considering the large proportion of public budgets spent on education, the scant

attention paid to producing reliable, detailed and timely statistics on the activities of this sector is quite surprising.

One result of the low priority given to education statistics is naturally a correspondingly low score as regards the allocation of resources to the collection and analysis of this type of statistic. This has meant a lack of staff and equipment. Furthermore, it has often led to education statisticians having poorer working conditions in terms of pay and career possibilities than statisticians employed in more prestigious fields of statistics. The education sector has therefore often lost out in the competition for qualified staff with other sectors of the economy. This applies to the initial recruitment of qualified staff as well as to the sector's ability to retain staff trained on the job. Frequent transfer of qualified personnel to other sectors is a problem in many developng countries.

(iii) *Lack of equipment*

This aspect has already been mentioned above but it deserves a separate point. Basic equipment such as typewriters, duplicating machines and calculators is very often lacking, particularly at the local level. For example, for India, Saraf *et al.* say that:

> At the district level calculating machines or other mechanical aids are not available. Compilation of data is, therefore, mostly done manually at this level. Manual calculations not only result in delay in the compilation work but also contribute to the errors in calculations.
>
> (Saraf *et al.*, 1980, p. 174)

The use of computers for processing education statistics is often discussed. Such facilities are perhaps not a top priority at present in countries where the quality of the data available is as low as is presently the case in many developing countries. In such countries the returns on the investments which would be required to computerize education statistics would perhaps be higher if these resources were spent on the data-gathering side (e.g. on improving accuracy, detail and completeness of the data) rather than on speeding up the processing of the low quality statistics presently collected.

But, naturally, at some point it will be in the interest of most countries to start using electronic computers for storing and processing of education statistics. When this point is reached depends on national conditions and must be decided by each individual country in each case.

With respect to equipment, the report of the Regional Adviser on Education Statistics in Africa quoted above states that:

> . . . A radical improvement of educational statistics is called for, but this must primarily be achieved through concerted action to reinforce the responsible unit, both with staff and equipment. A few of the countries have experimented with computer processing of their primary school statistics but not with entirely positive results. Such processing alone does not solve all the problems for these small units and should be introduced only after careful planning of all the phases in data collection and compilation.
>
> (UNESCO, 1977, p. 9)

(iv) *Lack of co-operation and support*

Those responsible for education statistics in developing countries often feel a lack of support and co-operation from schools as well as from other units and agencies within the administration. At the school level there is often a lack of appreciation as regards the importance of the data collected through the annual questionnaires. Furthermore, experience shows that headmasters, often overloaded with administrative and pedagogical responsibilities, cannot always be expected to devote enough time and interest to the filling in of statistical questionnaires. Particular attention needs moreover to be given to the role of school inspectors, local education officers and heads of educational institutions in the process of statistical data collection. Close and trustful collaboration between the various links of the data-gathering chain is a necessary condition for improving the quality of education statistics.

Particular problems of co-operation are often experienced with private schools as well as with public agencies providing instruction in institutions not controlled by the Ministry of Education. It is important that all educational institutions be obliged by law to complete the annual school questionnaires.

(v) *Problems of communications*

Among the many special problems faced by those responsible for education statistics in developing countries we mention, finally, the

very poor means of communication in many countries. This complicates the transmission of questionnaires to and from the schools as well as the process of rectifying errors and the collection of supplementary information to fill in gaps. Few primary schools in rural areas have telephones, they may be in remote areas far from roads, etc. The importance of close co-operation between headmasters, school inspectors and local education officers is, of course, even more important under such conditions than in countries where the central unit can contact schools easily by telephone.

## NOTES

1. To illustrate this problem we quote the following paragraph from Saraf *et al.* (1980). In discussing the gaps and deficiencies of education statistics in India, they state that:

> A large number of institutions, which do not fulfill the criteria for recognition and grant-in-aid, have come into existence in metropolitan cities and other urban areas. Although not recognized, they cater to the educational needs of certain sections of population. In many of these institutions, the standard of facilities and instruction provided may be substantially higher than available in the institutions maintained with public funds. Although they do not figure in the official educational statistics, these institutions nevertheless fulfill the felt needs of the community. To the extent it ignores these institutions, the official statistics does not indicate the total enrolment effort of the community. The disregard of the enrolment in the privately maintained primary schools for instance, has tended to give a false idea of the burden the state has to bear in universalizing primary education (p. 179).

2. For example, in most states of the Indian Union, data collection and compilation is carried out at three different administrative levels before being submitted to the Union Government, i.e. sub-division or block level, district and state level. Two states (Haryana and Tamil Nadu) even have four levels of data collection and compilation, see Saraf *et al.* (1980, p. 168). Also less populous countries channel the statistics through several levels of administration. For example, in the Sudan the questionnaires for primary schools go from the schools to the council, then to the province, then to the region which, finally, submits the data to the Central Government (see Fredriksen, 1982).

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